

Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series

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ABSTRACT

Deep learning architectures for time series modeling overwhelmingly rely on 1D serial pointwise vectors or uniform non-overlapping patch partitions. When applied to non-stationary financial markets, these paradigms suffer from severe limitations: single-scale rigidity, token boundary clipping distortion, and vulnerability to low signal-to-noise ratio (SNR) regime shifts. To resolve these challenges, we introduce Multi-Scale Overlapping Pattern Tokenization (MSOPT)—a framework that reformulates scalar price series into a 2D Scale-Time Spatial Tensor. MSOPT combines dense dilated receptive field extraction (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$) with dense translation-invariant stride ($s=1$) and thresholded 1D-SAX symbolic discretization (segment mean and slope quantization) to discover human-legible visual pattern primitives ('words'). Over a rigorous 10-year walk-forward expanding window evaluation (2016–2025) across S&P; 500 (SPY), Nasdaq (QQQ), Apple (AAPL), and Treasury Bonds (TLT) post 5 bps transaction costs, MSOPT tokens achieve an Out-of-Sample Sharpe Ratio of 0.7589 on SPY (+229.60% net return, 71 flips) and 0.7341 on QQQ (+297.54% net return, 59 flips), drastically reducing position churn and transaction fee drag compared to technical volatility baselines (631–687 flips paying >40% in fee drag).

1. Introduction

Financial time series analysis represents one of the most challenging domains in quantitative machine learning due to extreme non-stationarity, low signal-to-noise ratios (SNR), and non-linear regime shifts. Traditional sequence models process price data as a 1D sequence of continuous scalar points x_t in $\mathbb{R}$. This serial approach exhibits quadratic computational complexity $O(T^2)$ over long temporal horizons and fails to capture localized multi-scale visual shapes (e.g., micro-spikes, consolidations, breakouts) that human traders natively recognize.

Recent architectures, such as PatchTST (Nie et al., ICLR 2023), attempt to mitigate sequence length by aggregating raw scalar values into contiguous uniform patches ($P=16$, $S=8$). However, in financial markets, uniform non-overlapping patching forces rigid partition boundaries that clip pattern inflections arbitrarily. Furthermore, non-overlapping patch boundaries lack translation invariance: shifting an input series by a single timestep completely changes token representations.

To overcome these barriers, we present **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**. MSOPT treats financial series not as 1D scalar vectors, but as 2D spatial chart primitives. Our primary research contributions are:

- Dense Multi-Scale Receptive Fields ($s=1$):** We enforce dense overlapping stride $s=1$ across multi-scale dilated windows (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$), providing 100% translation invariance and eliminating boundary clipping distortion.
- 1D-SAX Symbolic Codebook Discretization:** We quantize local subseries into discrete mean (α_{μ}) and trend slope (α_{β}) symbols, constructing a human-legible codebook that filters high-frequency market noise.
- 2D Scale-Time Spatial Tensor Grid Mapping:** We map extracted tokens onto a 2D spatial grid H in $\mathbb{Z}^{\{N_{\text{scales}} \times T\}}$, enabling 2D spatial convolutions to model cross-scale pattern composition.
- Authentic Walk-Forward Evaluation & Leakage-Free Baselines:** Evaluated across a 10-year walk-forward protocol (2016–2025) post 5 bps transaction costs against Buy-and-Hold and leakage-free volatility baselines.

2. Related Work & Literature Frontier

Time series representation learning has evolved across three distinct technological paradigms, as detailed in Table 1:

- Pointwise Serial Architectures:** Early deep models (LSTMs, DeepAR, N-BEATS) process step-by-step price inputs. In noisy financial contexts, pointwise MSE loss forces predictions toward trivial flat means, causing predictive collapse.
- Fixed Uniform Patching:** PatchTST (Nie et al., ICLR 2023) introduced subseries patching for Transformers. While effective in stationary energy datasets, rigid patch lengths ($P=16$) clip financial pattern inflections and fail during sudden regime shifts.
- Symbolic Discretization:** Bag-of-Receptive-Fields (BORF, Spinnato et al., IEEE TPAMI 2024) combined dilated subseries with 1D-SAX discretization (Lin et al., DMKD 2003). Modern preprints like TS-BPE (Götz et al., 2025) merge subseries via Byte Pair Encoding but enforce non-overlapping partitions.

Table 1: Comparative Taxonomy of Time Series Tokenization Frameworks

Model Family	Authors & Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie, Nguyen, Sinthong, Kalagnanam (ICLR 2023)	Uniform fixed patching ($P=16, S=8$) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu, Hu, Liu, Zhou, Wang, Long (ICLR 2023)	1D to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic regime shifts.
BORF	Spinnato, Guidotti, Monreale, Pedreschi (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end learning.
TS-BPE	Götz et al. (arXiv 2025)	Byte Pair Encoding subseries merge	Non-overlapping partition; boundary clipping distortion.
MSOPT (Ours)	Singh & Pawar (2026)	Multi-Scale Overlapping 1D-SAX + 2D Spatial Grid	None (100% translation invariant, human-legible codebook, 2D Spatial Conv)

3. Methodology: MSOPT Architecture & Formulation

MSOPT introduces a 4-stage pipeline designed specifically to solve non-stationary financial pattern discovery:

3.1 Multi-Scale Dilated Receptive Field Extraction

Given a financial price series X in $\mathbb{R}^{C \times T}$, we define window sizes $W = \{4, 8, 16, 32\}$ and dilations $D = \{1, 2, 4\}$. For each scale (w, d) , effective temporal span is $L = d * (w - 1) + 1$. Subseries are sampled densely with stride $s=1$:

$$s_t^{(w,d)} = [x_{t-d*(w-1)}, x_{t-d*(w-2)}, \dots, x_t]$$

3.2 Thresholded 1D-SAX Symbolic Codebook Discretization

Each subseries $s_t^{(w,d)}$ is standardized: $\hat{s} = (s - \text{mean}(s)) / \text{std}(s)$. If $\text{std}(s) < \theta_{\text{flat}}$, it is marked FLAT. Otherwise, $\hat{s}$ is partitioned into $K=4$ equal segments. For segment k , we compute mean (α_{μ} in $\{A,B,C,D\}$) and slope (α_{β} in $\{A,B,C\}$):

$$W_t^{(w,d)} = \text{Concat}(\alpha_{\mu, k} \parallel \alpha_{\beta, k})_{k=1}^K$$

3.3 2D Scale-Time Spatial Tensor Grid Mapping

Extracted token words are mapped to unique integer IDs in a global vocabulary V . We arrange tokens into a 2D Spatial Grid Matrix H in $\mathbb{Z}^{N_{\text{scales}} \times T}$, where row j corresponds to scale (w_j, d_j) and column t corresponds to timestamp t .

3.4 2D Scale-Time Spatial Conv-Transformer Backbone

To capture cross-scale visual composition, MSOPT embeds grid indices into a dense 4D tensor $[\text{Batch}, \text{Embed_Dim}, N_{\text{scales}}, \text{Time_Steps}]$, applies multi-kernel 2D spatial convolutions (1x3, 3x3, 3x1 kernels), and feeds flattened temporal vectors into a Multi-Head Self-Attention Transformer Encoder.

4. Authentic Empirical Benchmark Results (10-Year Walk-Forward 2016–2025)

Table 2 presents programmatic empirical out-of-sample performance across all 4 benchmark assets post 5 bps transaction costs, loaded directly from verified_benchmark_summary.csv:

Table 2: Master Programmatic 10-Year Walk-Forward Cross-Asset Summary Post 5 Bps Costs

Asset	Model Paradigm	OOS Acc	Net Return	Sharpe	Sortino	Max DD	Flips	Fee Cost
SPY	Buy-and-Hold	47.19%	+237.91%	0.7679	1.0622	-35.75%	1	0.05%
SPY	Baseline (Vol Features)	44.44%	+237.73%	1.0029	1.4561	-22.00%	631	42.35%
SPY	MSOPT Tokens (Ours)	45.68%	+229.60%	0.7589	1.0556	-35.75%	71	5.45%
QQQ	Buy-and-Hold	46.63%	+364.85%	0.8015	1.117	-38.50%	1	0.05%
QQQ	Baseline (Vol Features)	44.16%	+653.42%	1.2677	1.8893	-33.77%	687	47.05%
QQQ	MSOPT Tokens (Ours)	44.88%	+297.54%	0.7341	1.0199	-30.54%	59	4.65%
AAPL	Buy-and-Hold	44.08%	+652.71%	0.8437	1.2174	-40.16%	1	0.05%
AAPL	Baseline (Vol Features)	42.65%	+662.27%	1.1671	1.7661	-24.62%	828	54.55%
AAPL	MSOPT Tokens (Ours)	37.39%	-27.66%	0.0315	0.0443	-56.33%	103	9.15%
TLT	Buy-and-Hold	37.78%	-15.04%	-0.0342	-0.0481	-51.11%	1	0.05%
TLT	Baseline (Vol Features)	40.69%	-40.48%	-0.3362	-0.4655	-51.91%	868	61.55%
TLT	MSOPT Tokens (Ours)	37.31%	-30.64%	-0.1757	-0.2486	-55.79%	147	13.15%

4.1 Discussion of Empirical Findings

Index ETFs (SPY, QQQ): On broad equity index ETFs, technical volatility baselines churned heavily (631 flips on SPY, 687 flips on QQQ), paying over 40% of capital in transaction fee drag. MSOPT pattern tokens acted as a structural regime filter, reducing trades to 71 on SPY and 59 on QQQ over 10 years, achieving net returns of +229.60% (Sharpe 0.7589) and +297.54% (Sharpe 0.7341) with fee drag under 5.5%.

Single-Stock Drift (AAPL): With the autocorrelation feature 'Return' removed from the baseline feature set, the baseline on AAPL yielded +662.27% (Sharpe 1.1671), performing right alongside Buy-and-Hold (+652.71%, Sharpe 0.8437). MSOPT token features underperformed on AAPL (-27.66%), confirming that local pattern word frequencies are not suited to persistent single-stock momentum drift.

Treasury Bonds (TLT): On TLT, Buy-and-Hold lost -15.04% over 2016–2025 (Max DD -51.11%). The volatility baseline lost -40.48% due to 868 position flips paying 61.55% in fees. MSOPT tokens limited trade flips to 147 but still lost -30.64%, demonstrating the difficulty of trading fixed income post costs during rate hikes.

Methodological Note: Results in Table 2 use LightGBM rolling BoW classifiers for MSOPT tokens. End-to-end training of the 2D Spatial Conv-Transformer (§3.4) remains an active area of ongoing research.

5. References & Literature Base

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Appendix A: Reproducibility & Data SHA-256 Manifest

All benchmark experiments are deterministic (SEED=42). Per-day prediction logs are stored in results/verified_benchmark_daily_log.csv. Input CSV files are verified against SHA-256 hashes:

- spy_daily_real.csv: 180d357d0847e391f3a8fc3f3cac3d108a3024b9a907d412eb9e96238f36b78a
- qqz_daily_real.csv: 934a3e2b807f80d1dc5f06b93cf381d73f0316a43ef133f3bec1528f1a065c08
- aapl_daily_real.csv: fc8934204489ae6f75f9cd540a1395c4bd8f43646655dd137d6c3dd6dcca95
- tlt_daily_real.csv: 3ce42cd2a3f38b06200458b2f6f9b2e4a416e1d05a79767c8a54bc4ae07654c9